

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

---

---

## Device Specifications

Device Type: Controller  
Manufacturer: Bosch  
Firmware Version: N/A  
Model(s): DCN-CCUB2

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 3.01.0000-b010                         | 2.2.0                   |

## Version History

| Module Version | Date       | Notes                                                                                             |
|----------------|------------|---------------------------------------------------------------------------------------------------|
| 1_2_0_0        | 4/4/2019   | Ethernet: Fixed MicrophoneControl and DualDelegateMicrophoneControl status. Added ChairmanStatus. |
| 1_0_4_0        | 12/26/2018 | Added command DualDelegateMicrophoneControlLastActioned                                           |
| 1_0_3_0        | 7/31/2018  | Initial Version                                                                                   |

## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- For ID assignment, the device will automatically assign IDs to the Microphone units connected to the main unit. Chairman Microphone units will take precedence over Normal Microphone units. For example, if there are 2 chairman microphones, those will take Microphone 1 and 2 on the driver. The rest of the Microphone units will take Microphone 3 and after.
- DCN-CCUB2 must be licensed before using the module. To check if it is licensed or not, you can go to <https://license.boschsecurity.com/register/dcn.asp>, key in S/N and Ch.
- Only delegate mic can be added to speaker list or add to request speaker list. On/Off state may not always be available for all delegate mics.
- Please refer to “Bosch DCN Next Generation Data Brochure” for configuring microphones.
- Configuring the mic as chairman or delegate.

### Functions

- Headphone output level reduction to prevent acoustic feedback (active when listening to the floor and when the microphone is on). When unit is used in Dual Delegate mode, this function is available individually
  - Built-in loudspeaker speaker is muted when the microphone is on to prevent acoustic feedback
  - Unit can be used as a single delegate unit, as a dual delegate unit, as a chairman unit (DCN-DISBCM chairman buttons, to be ordered separately) or as a single delegate unit with auxiliary button
- Configuring the button as ‘microphone on/off’ or ‘request-to-speak’.
  - What the Red/Green/Yellow indicator in the mic represents.

### Functions

#### Controls and indicators

- Built-in, fold-away, flat-panel loudspeaker, automatically muted if the microphone is on
- Microphone ‘on/off’ or ‘request-to-speak’ button
- ‘Microphone on’ indicator at the top of the loudspeaker
- Tri-color indicator above the microphone button:
  - Red ‘Microphone on’ indicator
  - Green ‘Request-to-speak’ confirmation indicator
  - Yellow ‘VIP’ indicator. ‘VIP’ LED is lit when the delegate unit is part of the notebook, which is only available when PC control software is used

---

## Supported Classes and Examples

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                             |
| InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DCN-CCUB2')               |
| <b>SerialOverEthernetClass</b>                                                                 |
| InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DCN-CCUB2') |
| <b>EthernetClass</b>                                                                           |
| InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9451, Model='DCN-CCUB2')           |

---

## Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

| Command                                                                   | Value  |
|---------------------------------------------------------------------------|--------|
| <b>EnableReturnCode</b> <sup>1</sup>                                      | 'None' |
| # EnableReturnCode example<br>InterfaceName.Set('EnableReturnCode', None) |        |

<sup>1</sup> This command must be set each time the device has been initialized.

## Status Available (Serial)

MicrophoneStatus do not support the Update function and is triggered by unsolicited data from the device.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

| Command                                                                                                                                               | Value           | Value |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|
| MicrophoneStatus                                                                                                                                      | 'On'            | 'Off' |
| Qualifier Key                                                                                                                                         | Qualifier Value |       |
| 'Microphone'                                                                                                                                          | '1' – '245'     |       |
| Value = InterfaceName.ReadStatus('MicrophoneStatus', {'Microphone': '1'})<br>InterfaceName.SubscribeStatus('MicrophoneStatus', None, FeedbackHandler) |                 |       |

## Control Commands (Ethernet)

Format with Qualifier:

InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})

Format with Qualifier:

InterfaceName.Set(Command, Value)

|                                                                                                                                                     |                                |                                |                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------|-------------------------------------|
| Command<br>ClearRequestToSpeakListAndSpeakersList                                                                                                   | Value<br>'None'                |                                |                                     |
| # ClearRequestToSpeakListAndSpeakersList example<br>InterfaceName.Set('ClearRequestToSpeakListAndSpeakersList', None)                               |                                |                                |                                     |
| Command<br>DualDelegateMicrophoneControl                                                                                                            | Value<br>'On'                  | Value<br>'Off'                 | Value<br>'Add to Speakers List'     |
|                                                                                                                                                     | 'Remove from Speakers List'    | 'Add to Request To Speak List' | 'Remove from Request To Speak List' |
| Qualifier Key<br>'Dual Delegate Unit'                                                                                                               | Qualifier Value<br>'1' – '245' |                                |                                     |
| Qualifier Key<br>'L/R'                                                                                                                              | Qualifier Value<br>'Left'      | Qualifier Value<br>'Right'     |                                     |
| # DualDelegateMicrophoneControl example<br>InterfaceName.Set('DualDelegateMicrophoneControl', 'On', {'Dual Delegate Unit': '1'})<br>'L/R': 'Left'}) |                                |                                |                                     |
| Command<br>MicrophoneControl                                                                                                                        | Value<br>'On'                  | Value<br>'Off'                 | Value<br>'Add to Speakers List'     |
|                                                                                                                                                     | 'Remove from Speakers List'    | 'Add to Request To Speak List' | 'Remove from Request To Speak List' |
| Qualifier Key<br>'Microphone'                                                                                                                       | Qualifier Value<br>'1' – '245' |                                |                                     |
| # MicrophoneControl example<br>InterfaceName.Set('MicrophoneControl', 'On', {'Microphone': '1'})                                                    |                                |                                |                                     |
| Command<br>WirelessMicrophoneControl <sup>1</sup>                                                                                                   | Value<br>'On'                  | Value<br>'Off'                 | Value<br>'Add to Speakers List'     |
|                                                                                                                                                     | 'Remove from Speakers List'    | 'Add to Request To Speak List' | 'Remove from Request To Speak List' |
| Qualifier Key<br>'Microphone'                                                                                                                       | Qualifier Value<br>'1' – '245' |                                |                                     |
| # WirelessMicrophoneControl example<br>InterfaceName.Set('WirelessMicrophoneControl', 'On', {'Microphone': '1'})                                    |                                |                                |                                     |

<sup>1</sup> The permitted state transitions and the state transitions initiated by the microphone button depend on the Microphone Mode setting on the device and the permitted number of Speakers allowed.

'On' will turn On the mic and add it to the Speakers List if there are vacancies.

'Off' will turn Off the mic and remove it from the Speakers List and the Request to Speak List.

'Add to Speakers List' will add the mic to the Speakers List if there are vacancies.

'Remove from Speakers List' will remove the mic to the Speakers List if it is in the Speakers List

'Add to Request To Speak List' will add the mic to the Request To Speak List

'Remove from Request To Speak List' will remove the mic to the Request To Speak List if it is in the Request To Speak List.

**Status Available (Ethernet)**

For RequestToSpeakList and SpeakersList, call Update to receive the latest status. ConnectionStatus, DualDelegateMicrophoneControl, MicrophoneControl, MicrophoneControlLastActioned, WirelessMicrophoneControl, and WirelessMicrophoneControlLastActioned do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

**Format with Qualifier:**

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

```
FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

**Format without Qualifier:**

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                                        |                             |                                |                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------|-------------------------------------|
| Command                                                                                                                                                                                                | Value                       | Value                          |                                     |
| ChairmanStatus                                                                                                                                                                                         | 'On'                        | 'Off'                          |                                     |
| Value = InterfaceName.ReadStatus('ChairmanStatus')<br>InterfaceName.SubscribeStatus('ChairmanStatus', None, FeedbackHandler)                                                                           |                             |                                |                                     |
| Command                                                                                                                                                                                                | Value                       | Value                          | Value                               |
| DualDelegateMicrophoneControl                                                                                                                                                                          | 'On'                        | 'Off'                          | 'Add to Speakers List'              |
|                                                                                                                                                                                                        | 'Remove from Speakers List' | 'Add to Request To Speak List' | 'Remove from Request To Speak List' |
| Qualifier Key                                                                                                                                                                                          | Qualifier Value             |                                |                                     |
| 'Dual Delegate Unit'                                                                                                                                                                                   | '1' – '245'                 |                                |                                     |
| Qualifier Key                                                                                                                                                                                          | Qualifier Value             | Qualifier Value                |                                     |
| 'L/R'                                                                                                                                                                                                  | 'Left'                      | 'Right'                        |                                     |
| Value = InterfaceName.ReadStatus('DualDelegateMicrophoneControl', {'Dual Delegate Unit': '1', 'L/R': 'Left'})<br>InterfaceName.SubscribeStatus('DualDelegateMicrophoneControl', None, FeedbackHandler) |                             |                                |                                     |
| Command                                                                                                                                                                                                | Value                       |                                |                                     |
| DualDelegateMicrophoneControlLastActioned                                                                                                                                                              | '1' – '245'                 |                                |                                     |
| Qualifier Key                                                                                                                                                                                          | Qualifier Value             | Qualifier Value                |                                     |
| 'L/R'                                                                                                                                                                                                  | 'Left'                      | 'Right'                        |                                     |
| Value = InterfaceName.ReadStatus('DualDelegateMicrophoneControlLastActioned', {'L/R': 'Left'})<br>InterfaceName.SubscribeStatus('DualDelegateMicrophoneControlLastActioned', None, FeedbackHandler)    |                             |                                |                                     |
| Command                                                                                                                                                                                                | Value                       | Value                          | Value                               |
| MicrophoneControl                                                                                                                                                                                      | 'On'                        | 'Off'                          | 'Add to Speakers List'              |
|                                                                                                                                                                                                        | 'Remove from Speakers List' | 'Add to Request To Speak List' | 'Remove from Request To Speak List' |
|                                                                                                                                                                                                        | 'Request To Speak'          |                                |                                     |
| Qualifier Key                                                                                                                                                                                          | Qualifier Value             |                                |                                     |
| 'Microphone'                                                                                                                                                                                           | '1' – '245'                 |                                |                                     |
| Value = InterfaceName.ReadStatus('MicrophoneControl', {'Microphone': '1'})<br>InterfaceName.SubscribeStatus('MicrophoneControl', None, FeedbackHandler)                                                |                             |                                |                                     |

## Global Scripter Module Communication Sheet

Revision: 4/4/2019

|                                                                                                                                                                                                                                                    |                                |                                |                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------|-------------------------------------|
| Command<br>MicrophoneControlLastActioned                                                                                                                                                                                                           | Value<br>'1' – '245'           |                                |                                     |
| Value = InterfaceName.ReadStatus('MicrophoneControlLastActioned')<br>InterfaceName.SubscribeStatus('MicrophoneControlLastActioned', None, FeedbackHandler)                                                                                         |                                |                                |                                     |
| Command<br>RequestToSpeakList                                                                                                                                                                                                                      | Value<br>'0' – '245'           |                                |                                     |
| Qualifier Key<br>'Index'                                                                                                                                                                                                                           | Qualifier Value<br>'1' – '25'  |                                |                                     |
| # RequestToSpeakList example<br>InterfaceName.Update('RequestToSpeakList', {'Index': '1'})<br>Value = InterfaceName.ReadStatus('RequestToSpeakList', {'Index': '1'})<br>InterfaceName.SubscribeStatus('RequestToSpeakList', None, FeedbackHandler) |                                |                                |                                     |
| Command<br>SpeakersList                                                                                                                                                                                                                            | Value<br>'0' – '245'           |                                |                                     |
| Qualifier Key<br>'Index'                                                                                                                                                                                                                           | Qualifier Value<br>'1' – '25'  |                                |                                     |
| # SpeakersList example<br>InterfaceName.Update('SpeakersList', {'Index': '1'})<br>Value = InterfaceName.ReadStatus('SpeakersList', {'Index': '1'})<br>InterfaceName.SubscribeStatus('SpeakersList', None, FeedbackHandler)                         |                                |                                |                                     |
| Command<br>WirelessMicrophoneControl                                                                                                                                                                                                               | Value<br>'On'                  | Value<br>'Off'                 | Value<br>'Add to Speakers List'     |
|                                                                                                                                                                                                                                                    | 'Remove from Speakers List'    | 'Add to Request To Speak List' | 'Remove from Request To Speak List' |
| Qualifier Key<br>'Microphone'                                                                                                                                                                                                                      | Qualifier Value<br>'1' – '245' |                                |                                     |
| Value = InterfaceName.ReadStatus('WirelessMicrophoneControl', {'Microphone': '1'})<br>InterfaceName.SubscribeStatus('WirelessMicrophoneControl', None, FeedbackHandler)                                                                            |                                |                                |                                     |
| Command<br>WirelessMicrophoneControlLastActioned                                                                                                                                                                                                   | Value<br>'1' – '245'           |                                |                                     |
| Value = InterfaceName.ReadStatus('WirelessMicrophoneControlLastActioned')<br>InterfaceName.SubscribeStatus('WirelessMicrophoneControlLastActioned', None, FeedbackHandler)                                                                         |                                |                                |                                     |



---

## Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

## Notes for the Device

---

## Serial communication

**Port Type:** RS-232

**Baud Rate:** 9600

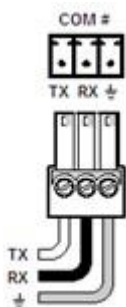
**Data Bits:** 8

**Parity:** None

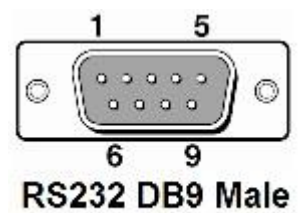
**Stop Bits:** One

**Flow Control:** None

## Pin Assignments Diagram



| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| TxD    |            | 2   | TxD    |
| RxD    |            | 3   | RxD    |
| GND    |            | 5   | GND    |



---

---

## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                            |          |
|----------------------------|----------|
| <b>Port Type:</b>          | Ethernet |
| <b>Default Port:</b>       | 9451     |
| <b>Logon Credentials</b>   | No       |
| <b>Supported:</b>          |          |
| <b>Multi-Connection</b>    | Yes      |
| <b>Capabilities:</b>       |          |
| <b>Port Changeability:</b> | No       |

---

---

## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device

---

## Appendix A. Set Commands(Serial)

|                         |                                    |
|-------------------------|------------------------------------|
| Enable Return Code None | \x1B\$\x0D\x03\x02\x00\x1E\x00\x90 |
|-------------------------|------------------------------------|



Rev. B1

[illegible]

## Appendix D. Update Commands (Ethernet)

|                                       |                                                             |
|---------------------------------------|-------------------------------------------------------------|
| <b>Heart Beat</b>                     | 'pD\x00\x10\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00         |
| <b>Request To Speak List Index 1</b>  | \x03%\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00%\x00 |
| <b>Request To Speak List Index 25</b> | \x03%\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00%\x00 |
| <b>Speakers List Index 1</b>          | \x03.\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00.\x00     |
| <b>Speakers List Index 25</b>         | \x03.\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00.\x00     |